SPE-228997-MS

# Optimizing Reservoir Performance Through Advanced Multistage Completion Techniques for Rigless Restimulation in North Kuwait Gas Field

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.



## Abstract

This paper evaluates the potential of restimulation of a well completed with multistage completions (MSC), highlighting the techniques that address production decline in wells while maximizing remaining reservoir potential through re-stimulation treatments in an existing well. This approach of selective re-stimulation is more challenging in Monobore cemented liners due to potential accessibility limitations, isolation of existing multiple perforations intervals particularly in highly deviated wells. These challenges often waive potential stimulation opportunities or limit stimulation efficiency of long perforation intervals in high permeability contrast heterogeneous carbonate reservoir.

With advancements in intervention strategies for the Jurassic gas field development, integrating advanced completions solutions has become a priority to improve productivity in tight carbonate reservoirs. The results support North Kuwait's Jurassic Gas strategy expanding its production capacity as the only producer of non-associated gas fields in Kuwait by demonstrating the improvement in well performance through innovative selective restimulation.

55 wells in the North Kuwait Jurassic fields were completed with ball-drop MSC in the North Kuwait gas fields. These wells, regardless of declining production rates still present significant reservoir potential. A key improvement includes using multi cycle re-closeable Frac sleeves and hydraulically activated shifting tools to facilitate temporary isolation and selective restimulation of targeted reservoir zones in a rigless approach. Real-time Fiber optic coiled tubing was deployed in some wells to provide real time monitoring of critical downhole parameters such as latching profile shifting and zonal isolation to verify closure/opening of frac-ports. Production Logging Tools (PLT) were run post-restimulation in some wells to assess the zonal contribution and validate the effectiveness of the restimulation treatments. Techniques such as acid and $CO_2$ foam fracturing were applied to improve selective restimulation operations. The case history of a well utilizing these approaches are presented highlighting operational challenges and outcomes.

Results from the restimulated MSC well show enhanced production, with targeted stimulation connecting natural fractures, closing depleted zones, restimulating other zones in addition to the reduction of the Frac plugs and reperforation usually required for restimulation. As a lesson learned the completion system was

redesigned and qualified to enhance pressure ratings after reclosing, ensuring that fracture design and pumping parameters during selective restimulation were not limited by the completion's pressure rating.

This paper introduces an integrated approach to rigless open-hole multistage restimulation operations, combining advanced intervention methods, real-time fiber optic monitoring, PLT evaluations and novel restimulation techniques.

## Introduction to North Kuwait Jurassic Reservoirs

The North Kuwait Jurassic Gas (NKJG) assets are currently being developed and managed by the Kuwait Oil Company (KOC). KOC is a subsidiary of Kuwait Petroleum Company, a fully owned national oil company of the State of Kuwait. NKJG assets encountered producible hydrocarbons at more than 14,000 ft TVDss – 15,000 ft TVDss in carbonate sequences beneath a salt structure in Sabriyah (SA), Raudhatain (RA), North-West Raudhatain (NWRA), Umm Niqqa (UN), Bahrah (BH), Dhabi (DH), Liyah (LH), Abdali (AD), and Sudaira (SUD) fields. The reservoirs cover an area of approximately 1,700sq. km.

The main reservoirs identified were the Middle Marrat carbonates (primary objective) and the Najmah Limestone together with a potential unconventional resource, an Organic Carbonaceous Shale in Najmah (OCS) as the secondary targets. The reservoirs are characterized by extensive natural fractures, which are currently being enhanced by stimulation to improve productivity. The initial pressure of the Middle Marrat reservoirs was approximately in the range of 11-12,000 psi. The reservoir temperature ranges from 260 – 300 deg F, with the Gotnia salt diaper creating a small change in the geothermal gradient. By the end of 2008, some 30 successful wells testing gas or volatile oil had been completed. At that stage, it was realized that some wells were capable of high production rates associated with a direct connection to extensive fracture networks. The Marrat reservoirs are interbedded with anhydrites and Najmah OCS related (source rock) layers which act as seals. The primary reservoir drive mechanism is depletion drive, subsequently, by 2013 the reservoir pressure had declined to 8,000/9,000 psi. The current reservoir pressure range is around 4,000-5,000 psi across the primary and most depleted sections. The Middle Marrat has an average permeability of about 10 mD and a vertical height of about 800ft. The permeability height function of Middle Marrat made it an attractive target and hence the majority of the NKJG field development has been focused in the Middle Marrat. An overview map of the asset is given in Figure *1*.

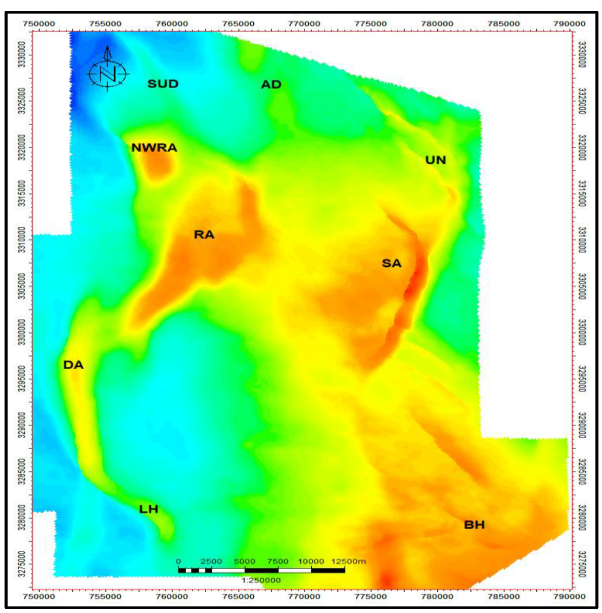


**Figure 1—Overview map of the NK Jurassic Gas Asset Fields**

Reservoir fluid characteristics constitute a combination of volatile oil and rich gas condensate with a dew point of 4,500 psi. The reservoir is sour, and the H2S concentration ranges from 3% to an excess of 10% in some fields while the CO2 is between 3 and 5%. The presence of produced water and insitu CO2 creates problems of corrosion and CO2-induced pitting in the carbon steel tubing. Formation water and CO2 created concerns with regards to tubing corrosion and surface handling issue. Recently, formation water production has led to outflow problems, including liquid loading and slugging. The lower completion of the well along the production path are made of corrosion resistant alloy (CRA) while the upper completion of the production conduit remains as the carbon steel to mitigate corrosion while preventing over-expenditure due to high cost CRA material.

The appraisal and development of the NKJG reservoirs offer challenges related to lateral variations in reservoir quality, tight to very tight reservoirs and natural fracturing to a varying degree. The presence of open, connected fractures is one of the key elements to achieve successful development.

The North Kuwait carbonate reservoirs are characterized by large spatial variations in fracture intensity which represents one of the key challenges for effectiveness of stimulation treatments. The presence of open, connected fractures can be a key parameter contributing to the success or failure of a well during completion and stimulation. While the presence of fractures is favorable to the productivity of the well, the presence of fracture corridors enhances the profile of a well after stimulation. However, on the downside, the fracture network can provide a channel to excessive water production after water breakthrough occurs. Detailed Fracture characterization using borehole image logs, Coriolis data from mud logs and modelling studies have been used extensively to support both appraisal and development strategies of these fractured reservoirs,

The geometry of the wells in NKJG fields are a mixture of vertical, deviated and horizontal wells. A 5″ or 4-1/2″ liner is run in the primary target in Marrat, and a 7-5/8″ liner is installed against the secondary target in the Najmah sequence. The reservoirs are typically acidized, hydraulic proppant fracture are placed in some wells in the secondary target in the Najmah OCS.

During the early stages of field development, the primary stimulation techniques were high-rate matrix acidization (HRMA) and the acid fracturing for areas with low matrix permeability. Several acid fracturing fluids, such as 28% HCl, gelled acid, emulsified acid, organic acid, 15% HCl and retarded acid systems were applied to achieve optimum fracture length and conductivity. As hydrocarbons recovered under depletion drive it was observed that there is reservoir pressure decline and differential depletion in some sectors of the RA and NWRA fields. This reservoir pressure decline has impacted the stimulation treatments and the ability to connect the matrix to the natural fracture networks in the depleted zones. The average initial production rates were approximately around 10 MMscf/d at a tubing head pressure of 4500psi. Currently, the average well rate is about 3 MMscf/d at average tubing head pressures of 800-1,200 psi. To enhance the effectiveness of the acid stimulation treatments and overcome the challenges due to differential depletion and reservoir heterogeneity of the Marrat reservoirs, the team decided to conduct a trial with energized CO2 acid fracture stimulation treatments in the Field RA and North-West Field RA (NWRA) fields (Danah et Al. 2022)

## Marrat Structure

The Jurassic Marrat formation is the main producer for NKJG. It is a heterogeneous tight carbonate reservoir, which lies below the Sargelu limestone and Dhruma shale. The Marrat includes Lower Marrat, Middle Marrat and Upper Marrat subunits or formations, as shown in Figure *2*. The Middle Marrat is the mature reservoir within the Marrat reservoirs, with about 1,500 ft total thickness across the field variable in different parts of the field. The formation is buried at vertical depths of ~ 14,000 to15,700 ft with the original reservoir pressure ranged from 9,000 – 11,000 psi with minimum horizontal stress gradient around 0.9 psi/ft. The reservoir pressure continues to drop as the depletion drive is chosen as the primary production mechanism with more wells drilled and produced.

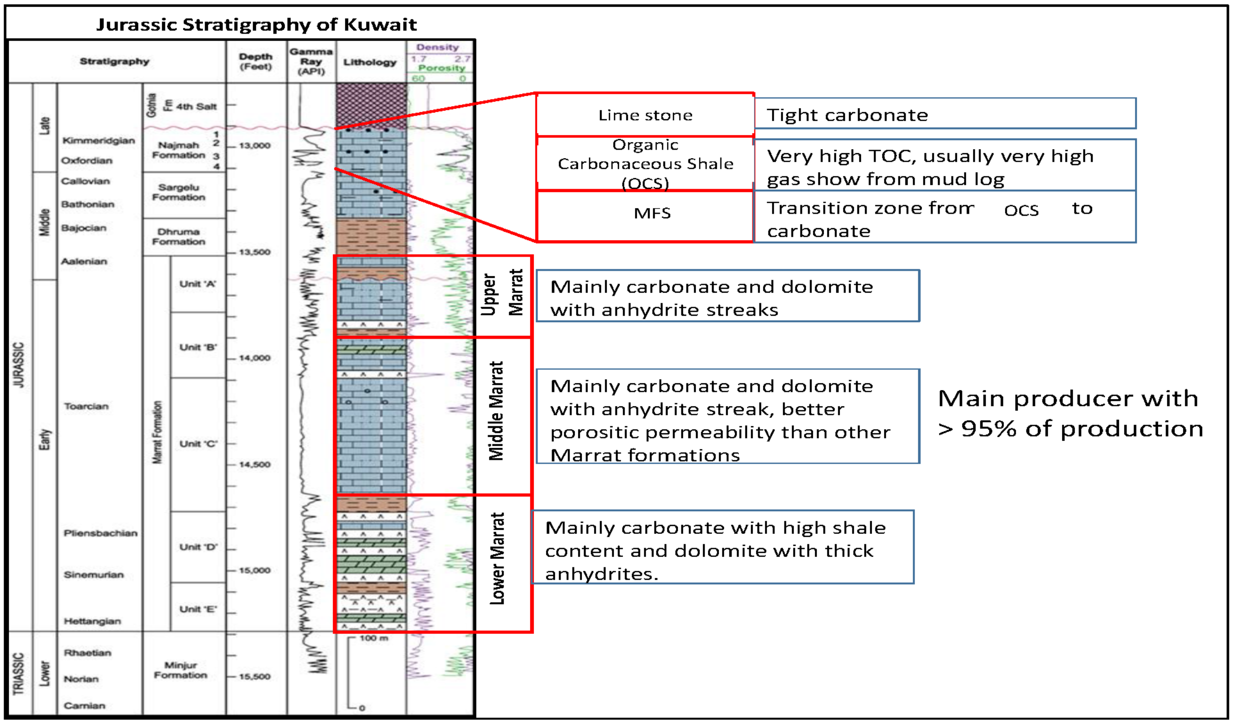


**Figure 2—Stratigraphic column highlighting the Jurassic Marrat formation.**

The formation is divided into three main reservoir sections: Lower, Middle, and Upper Marrat, with each unit further containing several thinner sublayers. The Middle Marrat is located above the Lower Marrat and is the main contributor to the production in the Marrat formation. It is a mainly dolomitic limestone formation and consisting of several flow zones with relatively higher permeability and porosity. The average permeability ranges from 0.11 to 100 mD and varies in the different flow zones. Several subunits within Middle Marrat are dolomitized grain stone facies with better porosity and permeability than the interbedded limestone.

Due to the high Jurassic carbonate reservoir tightness, permeability contrast among different flow units and dual permeability effect (matrix and natural fractures), well productivity significantly depends on the effectiveness of subsequent stimulation or re-stimulation treatments of such complex heterogeneous reservoirs to improve productivity and connect the natural fractures when present. Figure 3 illustrates the typical log showing high permeability contrast among reservoir layers.

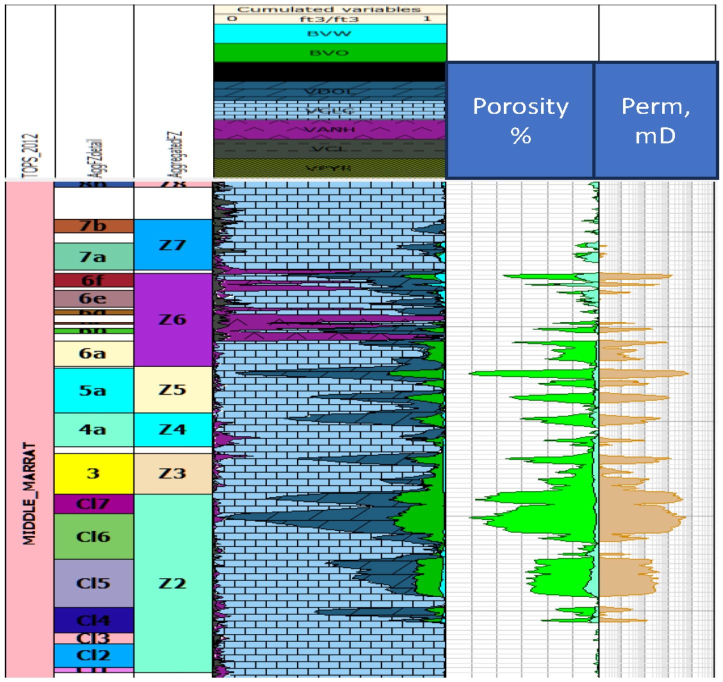


**Figure 3—Typical log showing high permeability contrast among reservoir layers**

## Jurassic Gas Completion History

The main challenge of North Kuwait Jurassic Gas HPHT deep reservoirs is that they consist of two major reservoir sections (mainly Najmah and Middle Marrat) with different pore pressures, rock and fluid properties. In the early stages of completing these reservoirs, tapered completion was commonly used. This involved drilling a 9.25-in open hole section across the first reservoir with a 7-5/8″ production liner, followed by drilling a 6″ open hole and installing a 5″ liner across the second reservoir section. Finally, the liners are cemented and the well is completed with a 3.5″ production string to the surface (Figure *4*, left-hand side).

The main downside of a tapered type of completion is that it requires liner cementation, which can damage the near-wellbore natural fractures and alter well productivity. Although tapered completion was suitable at the beginning of the production cycle for these reservoirs, it can be costly in the long term due to the need for a workover rig for any well intervention, such as zonal isolation, transfer, or remedial intervention (Al-Saeed et al., 2019).When preparing for a matrix acidizing campaign, a main concern is how effectively each zone at the near wellbore region can be isolated and treated independently. Following analysis and feasibility studies, three primary factors must be considered before proceeding: The effectiveness of isolation integrity, time and consumption cost, and the intervention risk involved. (Al-Othman et al, 2017).

Selective stimulation or any remedial well intervention was difficult due to the unavailability of downhole plugs suitable for such completions. Open hole completion with uncemented and slotted liners were tried as alternatives, but these designs did not lead to commercial and sustained well productivity and reservoir management. Barefoot completion is the most inexpensive method and is favorable for geomechanically stable reservoirs where natural fractures can be connected to the wellbore.

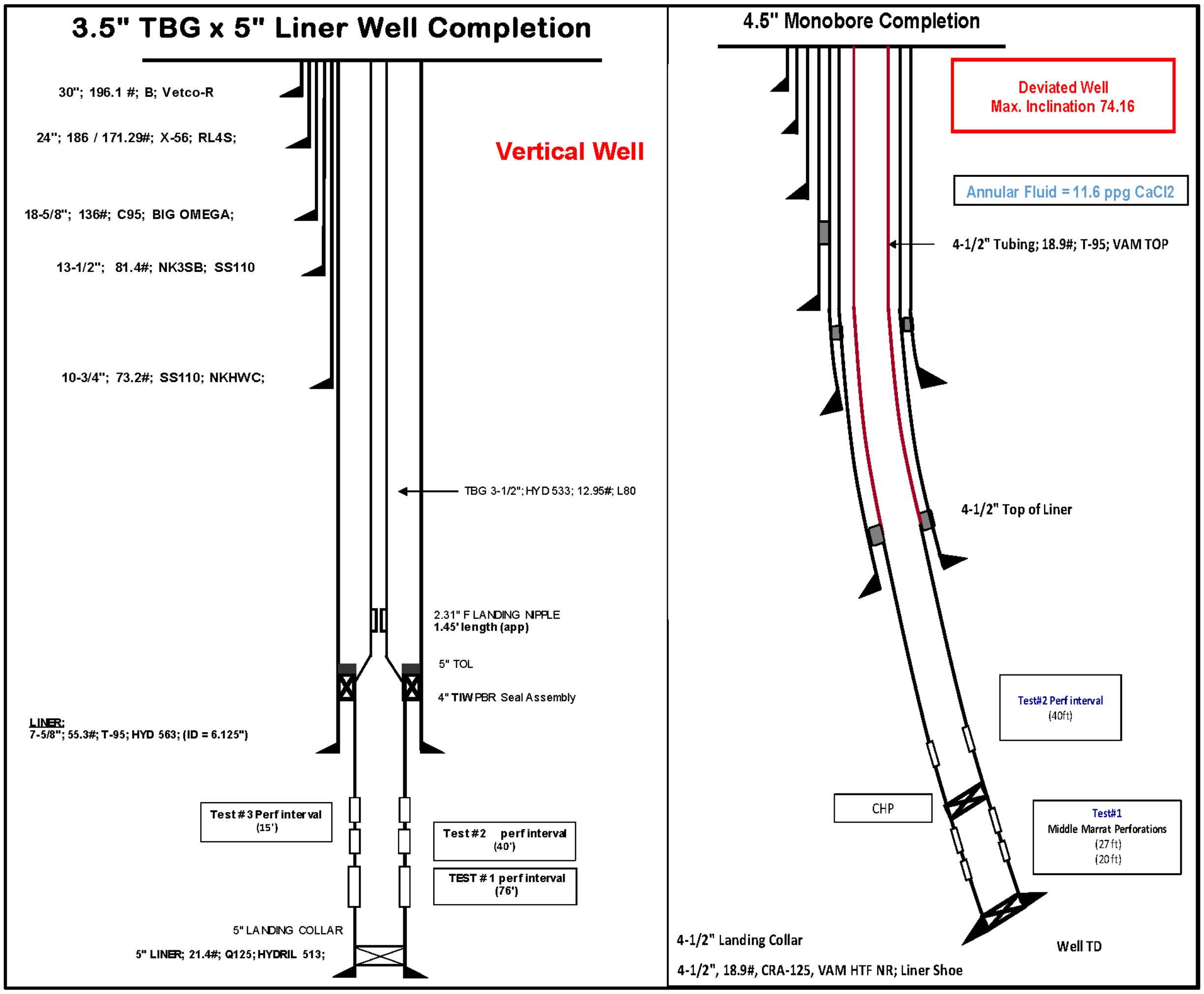


**Figure 4—Left-hand side shows a typical Jurassic Well completed with tapered 3.5″ tubing and 5″ cemented liner legacy completion, and right-hand side shows typical Jurassic Well completed with 4.5-in Monobore Completion Schematic**

High drawdown and pumping rates during high-rate matrix acidizing (HRMA) and fracturing can result in rock failure and wellbore collapse. To achieve complete coverage, a diversion technique may be used to redirect the treating fluid from one zone to another. There are two broad approaches to achieve diversions in stimulation treatments: mechanical isolation, and chemical diversion.

Consequently, the completion strategy was shifted to a monobore completion using a 4.5-in cemented liner across the main reservoir section, as opposed to a 5-in liner, and 4.5-in tubing string to surface instead of 3.5-in tubing (Figure *4*, right-hand side). This allowed for the installation of 4.5-in plugs, enabling rig-less mechanical isolation for plug and perf technique for selective stimulation of reservoir layers in a bottom-up approach using 15k bridge plugs to achieve mechanical fluid diversion.

Monobore 4.5-in cemented completions showed some accessibility limitations, especially in highly deviated wells (>60 deg) where wellbore cleaning challenges exist and could be considered sub-optimal due to the unavailability of a slim drilling string to clean inside a 4.5-in liner. This issue has adversely affected some wells' perforation and stimulation intervention plans, delaying well delivery to production. Additionally, cemented monobore completions could limit connectivity to any natural fractures in the reservoir, affecting overall well productivity potential, and requiring multiple well interventions for perforating long intervals (up to 650 ft net perforations) and subsequent isolation of planned stimulation stages (on average five stages per well). This adds operations challenges, resource consumption, mobilization logistic challenges, and excess time to deliver new wells to production.

## Evolution and Advances in Middle Marrat Multi Stage Completions

The evolution of multistage completions in North Kuwait's Middle Marrat reservoir represents a major step forward in addressing the dual challenges of production decline and complex reservoir behavior. KOC transitioned from traditional cemented completions to advanced open-hole multistage systems that enabled precise stimulation of natural fracture networks while offering greater control over zonal production. The customized system integrates multi-cycle closeable sleeves placed between open-hole packers, allowing up to 15 individual stimulation stages. Each sleeve is activated through a ball-drop mechanism, where progressively larger balls enable selective stimulation without intervention, reducing the risk and cost associated with wireline or coiled tubing operations. Figure 5 illustrates the differences between bullhead treatment, plug-and-perf operations and open hole ball-activated sleeve operations,

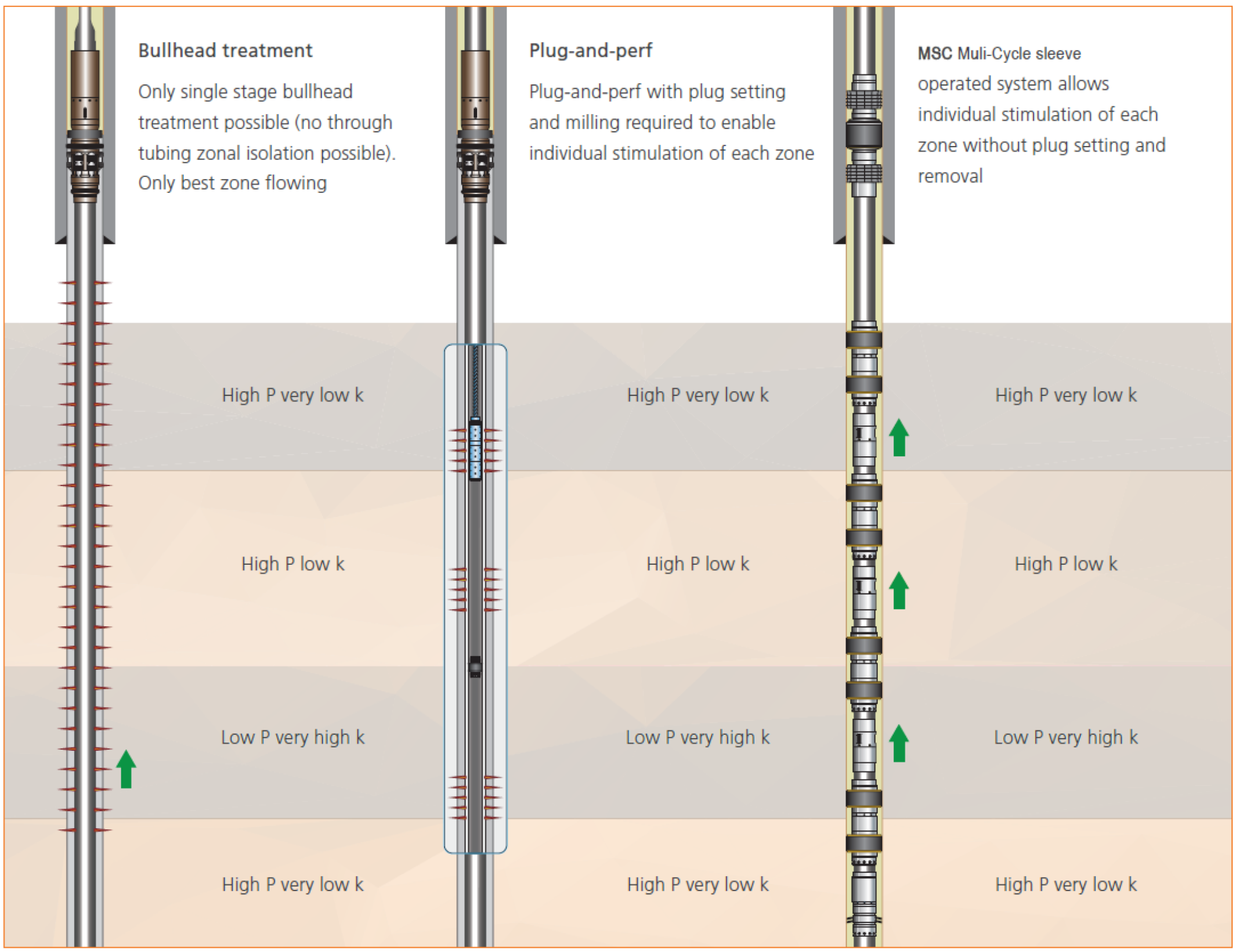


**Figure 5—Differences between bullhead treatment, plug-and-perf operations and open hole ball-activated sleeve operations**

To meet the demanding high-pressure, high-temperature (HPHT) environment of the Middle Marrat, the completion system utilized specialized technology and components constructed from corrosion-resistant alloys (CRA). These materials were specifically selected to endure differential pressures up to 15,000 psi and to withstand the acidic stimulation fluids used in carbonate acid fracturing treatments. Material compatibility was a critical design factor, especially given the long-term exposure to aggressive downhole conditions and the need for sleeve integrity over multiple opening and closing cycles. In later production phases, sleeves could be selectively reopened or closed using coiled tubing hydraulic shifting tools, allowing for targeted re-stimulation or the shutoff of unwanted fluid zones, such as water breakthroughs. This flexibility became increasingly vital as the field experienced production decline and increasing water cut, enabling dynamic reservoir management in response to change well performance.

## Business Challenges related to re-stimulation requirements

Initial Stimulation is typically performed using a bottoms-up approach, starting from the toe of the well with the upper stages in a closed and isolated state. These closed sleeves were engineered to withstand significant differential pressure via a primary isolation barrier before initial activation and during the stimulation of the lower zones, preserving the system's integrity while maintaining a monobore ID. The built-in re-closeable feature was meant only for water shut-off applications via a secondary barrier.

However, operational challenges emerged when KOC began exploring selective re-stimulation strategies —particularly when attempting bullhead acid treatments after previously milling all the ball seats followed by closing all sleeves and open the target sleeve for re-stimulation with no internal isolation between the stages. In this scenario, while selectively treating in the open sleeve, the other reclosed sleeves were exposed to high differential pressures from the surface down, a reversal from the original design where the secondary barrier is meant for water shut off. This led to the mechanical deformation of secondary internal sleeve components, compromising their ability to re-activate and, in some cases, intentionally limiting the maximum allowable surface treating pressure to protect them.

These challenges prompted the need for a complete redesign of the sleeve system, considering the growing requirement for re-stimulation flexibility without sacrificing the Monobore profile. The redesigned sleeves were engineered to maintain mechanical integrity under reversed differential loads after reclosing and enable multiple open-close cycles, ensuring future compatibility with bridge plug isolation, selective re-perforation, or coiled tubing intervention.

## Proposed Restimulation Solution in Multistage Completions

As a solution to the afore-mentioned challenges, the team has proposed the energized Carbon Dioxide (CO2) foam acid fracturing technique with 50% foam quality at surface. Energized CO2 fracturing is typically performed in depleted formations to enhance flowback of the fracturing fluid and to increase initial post-treatment production. The CO2 is pumped as the liquid phase, but it eventually becomes supercritical CO2 in the reservoir at downhole conditions. This is converted to gas phase in the wellbore during flowback and can help with clean up and to revive the well. It is also used to stimulate water-sensitive formations. Foamed CO2 minimizes the amount of fracturing fluid needed and limits fluid retention in the formation for enhanced production and efficient cleanup.

Conventional fracturing using water-based fluid systems alone have disadvantages such as utilization of large volumes of water, gel, complex chemicals, causing treatment damage to the reservoir induced gel filtrates, and by clay swelling and fluid residue. Fracturing with CO2 is a rising fracturing technique that cuts down on water-based fluids, including the acid itself, and the complex fracturing chemicals, which have been widely used in reservoirs with a permeability range of 0.1 to 10,000 mD.

CO2 energized acid fracturing stimulation treatments in North Kuwait Jurassic Gas operations have been applied effectively to mitigate the effects of reservoir depletion during acid fracturing operations. The application of CO2 has proven to be a significant game changer as an energy source to assist in the fluid recovery of well stimulation fluids. In terms of fracturing fluids, there are very limited applications for low permeability tight carbonate reservoirs due to complexities associated with the physical and mechanical properties of carbonate rocks and its interaction with the fracturing fluid. The advantages of utilizing super critical CO2 to energize the stimulation fluids has led to the reduction of potential formation damage normally associated with fracturing fluids and very rapid cleanup of the residual biproducts of the stimulation.

In North Kuwait Jurassic Gas fields, the reservoir depletion of the primary carbonate reservoir was the driving force and the key consideration for implementing a change in the stimulation technology. Addition of liquid $CO_2$ to HCl in significant volumes at high rate at surface, generating up to 50% foam quality by volume creates stable foamed emulsion which enables the HCl molecules to be retarded and penetrate much deeper than neat HCl itself would achieve. The liquid $CO_2$ can behave as a miscible phase, and can lighten up the reservoir fluids, thus potentially stripping the hydrocarbon from the rock which in return allowing acid to effectively react with the rock surface. This also creates a self-diversion effect, pushing the acid to etch new paths in the previously unstimulated sections of the dolomite. $CO_2$ miscibility with acid is further improved because it remains in the liquid or supercritical phase during injection into the reservoir and flows back in gas phase over a wide range of temperature and pressure allowing stimulation fluids to be recovered in a fast and safe manner (Danah et Al 2022). The benefits of $CO_2$ are

- Low tubular friction and high hydrostatic pressure (low surface injection pressure)
- Interfacial tension reduction
- High fluid expansion during flowback
- Long time energy preservation on flowback

## Implementation of Restimulation Campaign in Open hole MultiStage Completions (example)

### Case Study Well A

Well A is a deviated (54.7 deg) development well in the Eastern platform of the Raudhatain field drilled for the Middle Marrat primary target and Najmah-Sargelu as the secondary target with reached TD at 16,605ft (MD). The regional maximum horizontal stress direction is NE-SW, the well was drilled subparallel to minimum horizontal stress.

***Structural Petrophysics and Well Completions.*** Middle Marrat top was encountered at 14,784 ft TVDSS. Formation evaluation shows 6 flow zones (FZs), or zones of interest were identified. The well was drilled with an open-hole lateral length of approximately 1,265 ft completed by incorporating an advanced configuration of six stages of ball drop Multi-Stage Completion, using open hole mechanical packers to isolate between the stages. The frac ports were ball drop activated with the option of multiple open/close feature which enable the team to take selective decisions when it comes to stimulation and production selectively. The well completion and the composite log, as well as the multi-stage completion stages are highlighted in Figure *6* and Figure *7* respectively.

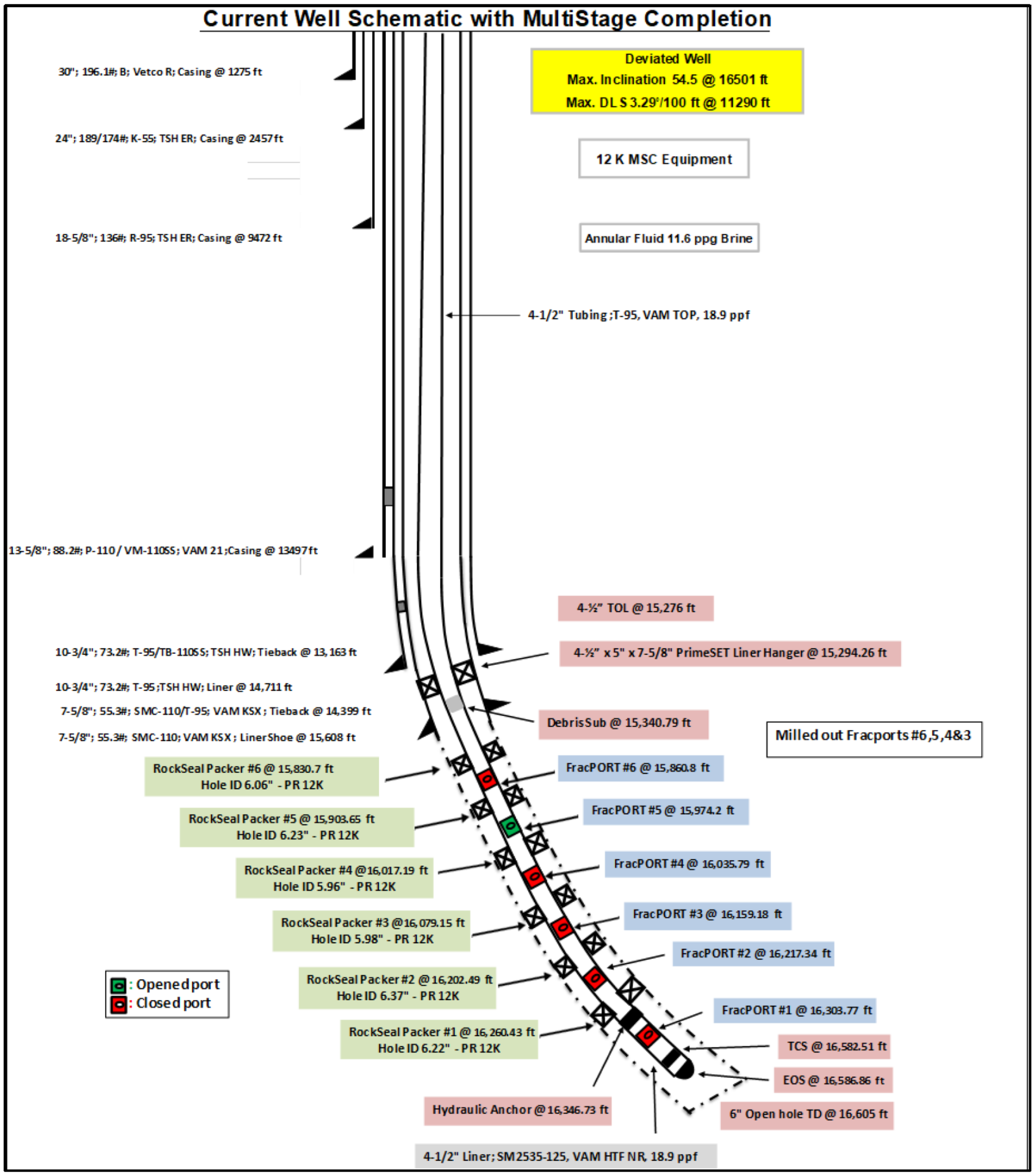


**Figure 6—Wellbore completion showing the status of Multi-Stage Open Hole completion in RA well**

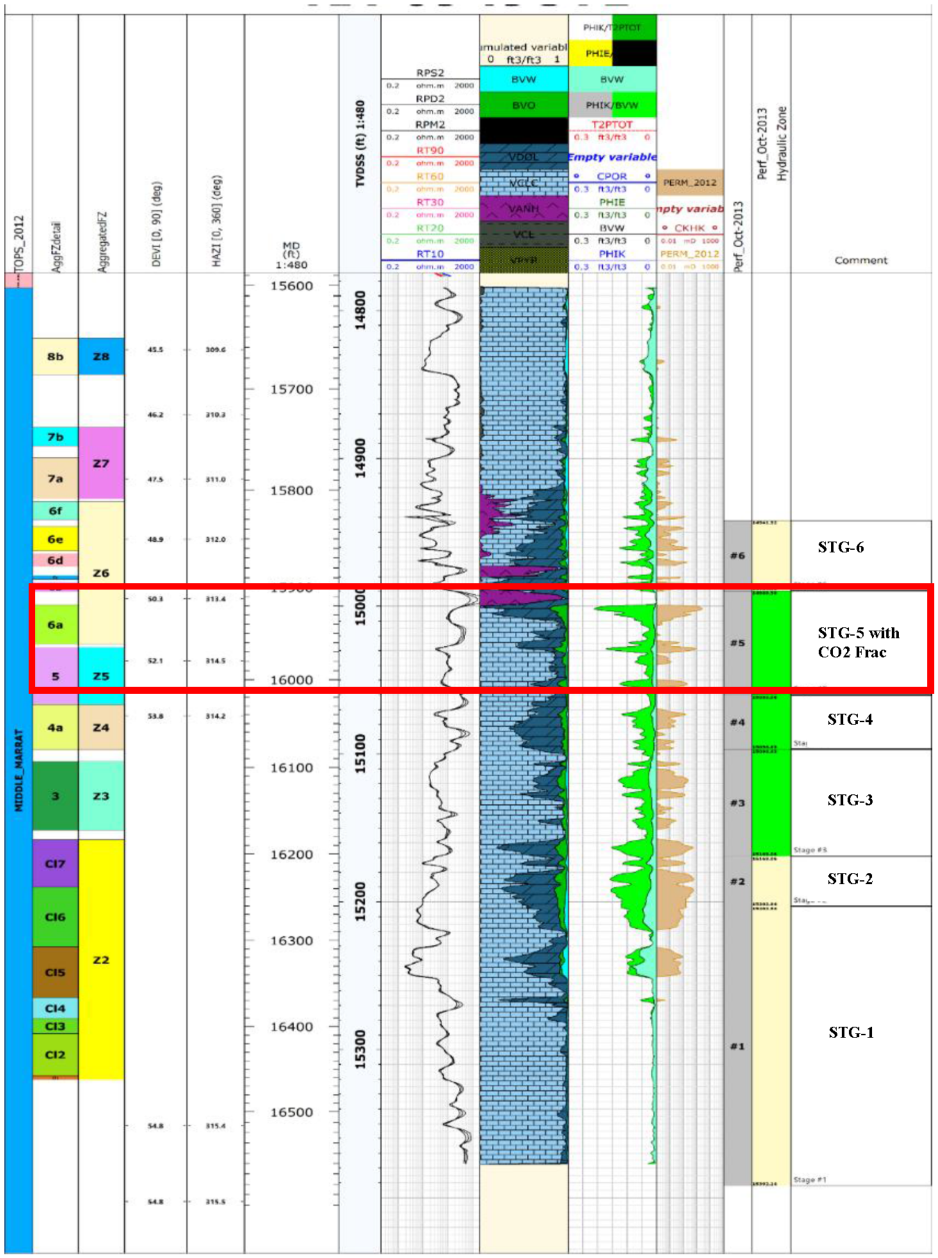


Figure 7—Composite Log of the well highlighting the Stage 5 as the zone of interest for re-stimulating with CO2 Fracturing

***Initial Stimulation and Production history.*** After completion, the well was initially stimulated using nitrified HRMA stimulation by bullheading at matrix injection rates for STG-3, STG-4 and STG-5 seperately,corrospending to Z#3, Z#4 and Z#5 respectively. Meanwhile STG-1, STG-2 and STG-6 were kept closed and not stimulated due to high risk of water breakthrough. Prior to the re-stimulation, the well produced at 1.42 MMscf/d at 452 psi on 32/64″ choke with 12% water cut per Figure 8 below. After sometime, the well experienced a significant wellhead pressure drop and production decline, thereafter the well stopped flowing.

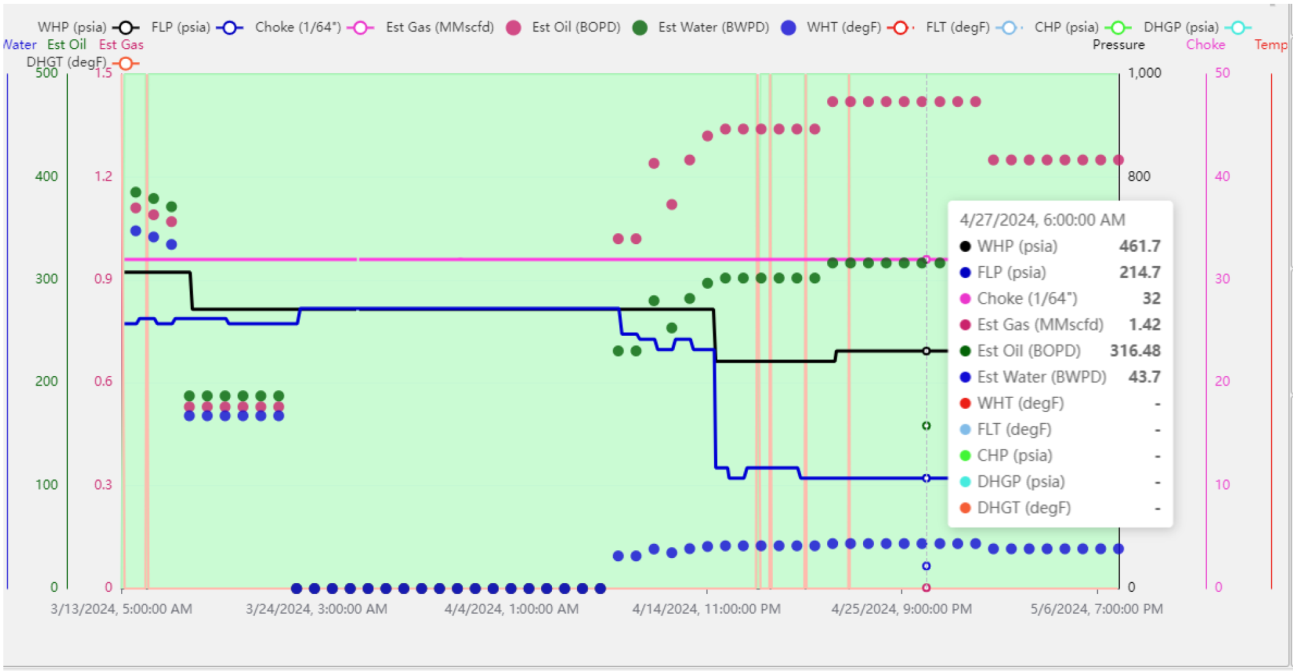


**Figure 8—RA Well's production history prior to re-stimulation**

***Re-stimulation Plan.*** The decision was taken by the team to re-stimulate stages of STG-5 using energized CO2 Acid Frac with 50% foam quality. NKGJ team developed a comprehensive plan involving multiple disciplines team including stimulation company to revive the well potential and revive the well into production.

The re-stimulation plan consisted of:

**Stage-5 Re-Stimulation with Energized CO2 Acid Frac**

Re-stimulation plan of stage#5 was as follows:

- Closing Stage 3 and Stage 4 fracports using shifting tool run on coil tubing unit
- Perform mini injection test on stage-5 and refine the main treatment schedule for the energized CO2 frac
- Perform energized CO2 acid frac for STG-5
- Flow back Stage 5 after acid soaking and test on different choke sizes to evaluate the performance of the zone.

***Frac Design.*** CO2-energized acid fracturing technology was established in NKJG since 2021 with the objective of improving the stimulation and hydraulic fracturing efficiency in the Jurassic Middle Marrat formation. CO2-foamed acid fracs have several advantages over other stimulation techniques in re-simulation of the non-producing wells

- CO2 foam bubble is an effective leak off control when treating stimulated zones, injection pressure tends to be minimum due to high leak off leaving reservoir unstimulated properly.
- CO2 is a miscible and non-damaging fluid blends in water and mixes with hydrocarbons.
- Pumped as a liquid and slightly heavier than water, leading to lower treating pressures due to heavier hydrostatic head.
- Effective in treating lower-pressured/partially depleted good KH carbonate reservoirs.

- Reduces water-based gels and overall frac-load volume by the percentage of $CO_2$ pumped in the fluid system (50% by volume is utilized in this case study well).
- Energizes the frac fluid and stays in solution until it heats up to gas. This property ensures the frac load recovery is achieved throughout the flowback.
- Eliminates the need to activate the well after the frac with CT/$N_2$ applications potentially saving time and money.
- $CO_2$ lightens up the heavier ends of the hydrocarbons due to its miscible properties. Hence, it helps with better hydrocarbon inflow.
- It creates a stable foam structure with the frac fluid, increasing the frac fluid viscosity hence has the potential to generate better frac geometry and higher "stimulated rock volume" or SRV.

***Fracturing Workflow.*** To plan the $CO_2$ foamed acid fracturing activity, standard design and execution sequences as well as the additional operational assessments and preparations were made. Special cryogenic $CO_2$ supply and heat exchanger equipment were provided. Service company pumping equipment specially designed to pump liquid sub-zero cryogenic $CO_2$ along with acid fracturing equipment to handle 15,000 psi surface pressure test and 13,500 psi working pressure were rigged up and utilized safely.

Workflow of the Process:

- Data Gathering
- Laboratory Solubility and fluid Compatibility Testing
- Synthetic Stress Log Generation
- Fracture Modelling
- Datafrac design
- Mainfrac design

Programming the execution plan, and coordination with Service Company to execute on-site testing of the $CO_2$ pumping equipment, including cool down and leak tests of the high-pressure pumps and manifolds in advance. This is essential since the treatments were pumped at ambient conditions reaching 45 / 50 deg C.

The most important pre-stimulation activity was the modeling of the $CO_2$ foamed fractures using a fracture simulator. Pseudo 3D models were generated for each Middle Marrat reservoir flow zone to be stimulated in the correct sequence to determine fracture half-length and fracture conductivity. The fracture model was used to determine the treatment volumes and the optimum acid to $CO_2$ ratio to provide enough diversion to the lower permeability matrix.

The execution of the datafrac and fluid calibration tests were key to the success of the main fracturing treatment. The team relied very heavily on geological understanding and calibration with dynamic data, especially where there are extensive connected fractures and fracture corridors which are typical of the RA field.

The fluid calibration and fluid efficiency tests were used to detect fluid leak off, impact of fracture networks on fracture height growth and effect of on stress variations on fracture impedance, fracture conductivity and diversion to low permeability matrix regions of the reservoir. The anhydrite layer in the Marrat formations has a strong impact on the pressure compartmentalization vertically. From the datafracs pumped and subsequent dipole sonic logs of the acid fracture, it has been observed that induced acid fracture cannot cross the interfaces between anhydrite beds. The $CO_2$ fractures were limited to flow zone sequences of the middle Marrat which are free of anhydrite layers. This learning was transferred from previous studies done on discrete fracture modelling. (Richard et al 2017, ref 6)

***Fracture Execution.***

- Datafrac execution and main frac design calibrations
- Main frac placement
- Post Fracture evaluation and production testing using on-site portable separate and multi-phase flow meter unit and a Green Burner to reduce flare gas and oil.

***DataFrac: DFIT (Diagnostic Fracture Injection Testing).*** This test is done to determine the leak off trend, closure gradient and fluid efficiency for design of the main frac treatment. It is done with the Newtonian fluid and at high rate. It is normally done with a volume of fluid that is large enough that most of the rock volume that is to be fractured will be contacted. After the test, the leak off was monitored, till pseudo radial flow regime to ascertain the reservoir properties. G-Function analysis was done to check the leak off trend, which showed that slight PDL in this formation is dominant, and rate control is crucial for achieving the fracture length required.

***Main Fracture Treatment.*** Based on all the testing results and analysis, the fracturing treatment design was further optimized, and rate, volumes and acid concentrations tweaked to facilitate high conductivity gain. Three alternate cycles of PAD, main-acid and diverter acid (Table 1) were performed to enhance the ramifications of wormhole in this depleted zone. Also, pumping rate was maintained to facilitate net pressure gain and overcome high leakoff trend.

**Table 1—Energized CO2 Acid Fracturing Treatment Pumping Schedule**

| Cycle # | Stage | Stage Type | Liquid Rate | Fluid Vol. BBL | Cum. Vol. BBL | CO2 Rate BPM | CO2 Vol. Tons | Cum. CO2 Tons | Total Rate Acid+CO2 BPM | Surface Foam Quality % | Time Min. | Fluid Type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | Pre-Flush | 10* | 100 | 100 | 10* | 16.2 | 16.2 | 20 | 50 | 10 | Mutual Solvent |
| Cycle# 1 | 2 | PAD-1 | 10* | 200 | 300 | 10* | 32.4 | 48.6 | 20 | 50 | 20 | 25# Low Visc XL |
| | 3 | 15% HCl Acid-1 | 10* | 100 | 400 | 10* | 16.2 | 64.9 | 20 | 50 | 10 | 15% HCl |
| | 4 | VES Diverte r-1 | 10* | 100 | 500 | 10* | 16.2 | 81.1 | 20 | 50 | 10 | 7.5% VES, 15% HCL |
| | 5 | PAD-2 | 10* | 200 | 700 | 10* | 32.4 | 113.5 | 20 | 50 | 20 | 25# Low Visc XL |
| Cycle# 2 | 6 | 15% HCl Acid-2 | 10* | 100 | 800 | 10* | 16.2 | 129.7 | 20 | 50 | 10 | 15% HCl |
| | 7 | VES Diverte r-2 | 10* | 100 | 900 | 10* | 16.2 | 145.9 | 20 | 50 | 10 | 7.5% VES, 15% HCL |
| Cycle# 3 | 8 | PAD-3 | 10* | 100 | 1000 | 10* | 16.2 | 162.2 | 20 | 50 | 10 | 25# Low Visc XL |
| | 9 | 15% HCl Acid-3 | 10* | 100 | 1100 | 10* | 16.2 | 178.4 | 20 | 50 | 10 | 15% HCL |
| | 10 | Post Flush | 10* | 100 | 1200 | 10* | 16.2 | 194.6 | 20 | 50 | 10 | Mutual Solvent |
| | 11 | CFA | 10* | 100 | 1300 | 10* | 16.2 | 210.8 | 20 | 50 | 10 | 15% HCL |
| | 12 | Over Flush 1 | 10* | 116 | 1416 | 10* | 18.8 | 229.6 | 20 | 50 | 11.6 | 2% KCl Brine |
| | 13 | Over Flush 2 | 5 | 100 | 1516 | 5 | 16.2 | 245.8 | 10 | 50 | 20 | 2% KCl Brine |

The main acid fracturing treatment design was based on the following criteria:

- Ability to achieve a long conductive fracture (half-length >500 ft, conductivity >1000 mD-ft)
- Ability to achieve confined fracture to avoid any possibility of breakthrough into a possible water-bearing zone below, several runs of fracture simulation were applied prior to the selection of an optimized treatment pumping schedule.
- Ability to flow back the well naturally without the need for coil tubing and nitrogen activation.

The optimal energized CO2 Acid fracturing schedule considers a total dosage of 1100 gals/ft with 50% Foam quality at surface (acid dosage of 550 gals/ft, 550 gals/ft liquid CO2). The acid ratio between straight acid and leak-off control acid as diverter with the volume of (50% Main Acid and 50% Leak-off control acid) respectively, PAD volume of 1:1 ratio to total acid volume. The proposed fluid system was optimized based on previous experiences within Middle Marrat in the RA field (B. Al-Enezi et Al.2024). Table *1* below shows the executed pumping schedule for STG-5.

***Fracture Simulation Design output.*** Figure *9* Shows fracturing a fracture half-length of more than 500 ft covering Z5 with an average conductivity of 3,341.9 mDft

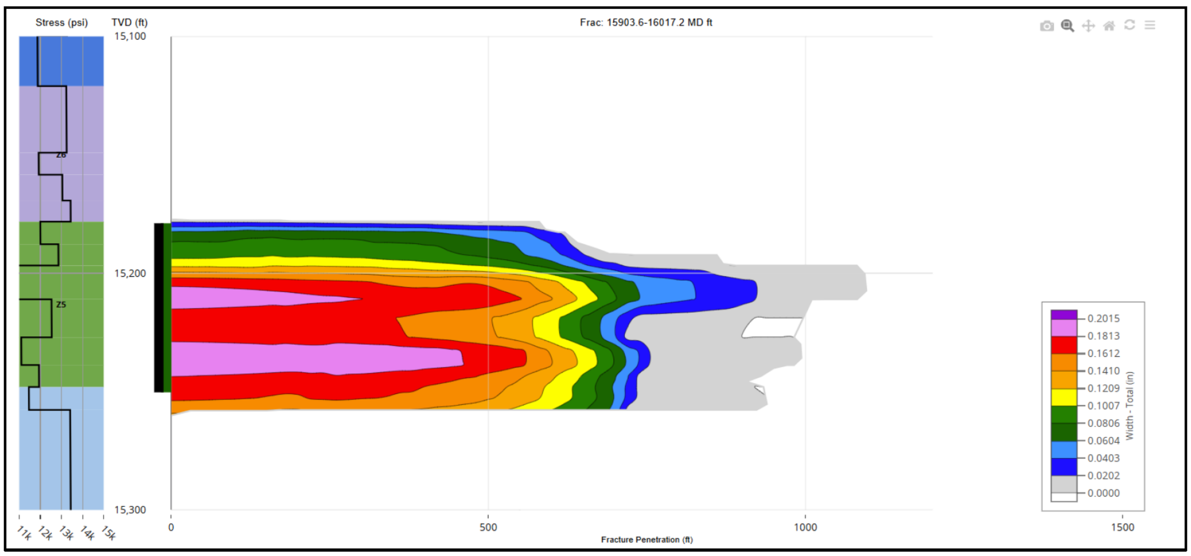


**Figure 9—Energized CO2 Acid Fracturing treatment simulation results for stage 5 corresponding to Z5.**

***Stage-5 Execution.*** As planned, operation began with STG-5 re-stimulation after closing frac port 3 and port 4 using coil tubing shifting tool, followed by the frac operation as follows:

- Performing hole fill and Step Rate Test (SRT)
- Analysis of the Step Rate Test (SRT)
- Perform the Main Treatment

## Main Treatment

As planned during the main treatment CO2 surface foam quality was maintained as 50% during the job. The maximum rate achieved was 22 bpm due to MASP limits of 7500 psi and friction pressure, as shown in Figure *10*. The PAD cycles were creating positive pressure when it's at perforation and this is because of the good leak off control created by the CO2 bubbles creating deep penetration inside the target zone as predicted by the fracture simulation in Figure *9*.

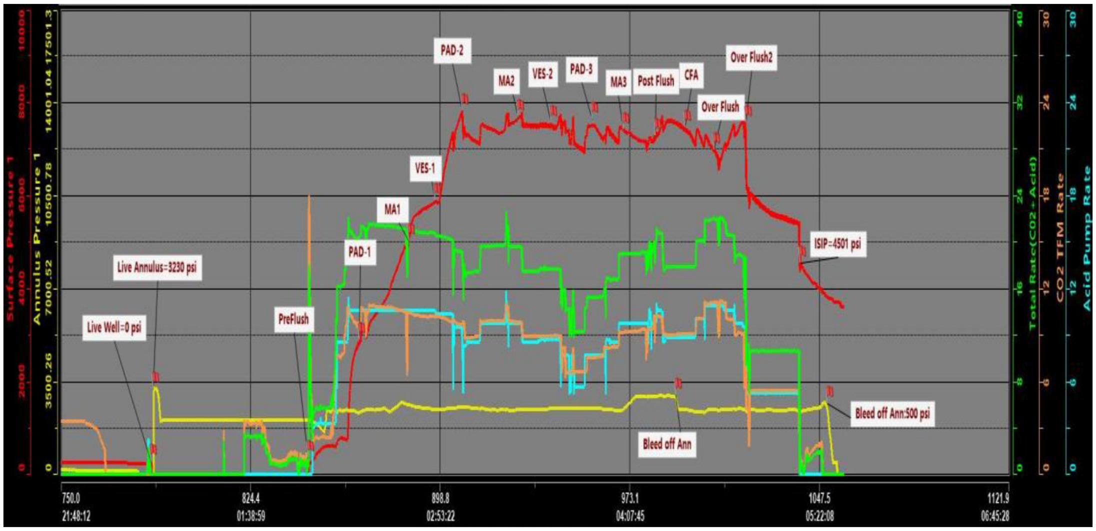


**Figure 10—STG-5 Energized CO2 Acid Fracturing main treatment pumping chart**

The Main Acid etched the created geometry more clearly in cycle 2 and 3. A total of 1516 bbls of fluids and 270 MT of liquid CO2 was injected in the Z5 formation successfully without any incident. The total CO2 dead volume considered for this stage was 40 MT to compensate for the venting, cooling down volumes. After the well was shut-in for several hours for fracture closure and acid reaction and soaking. Then it was flowed back at multiple choke sizes through the testing equipment.

## Results and Achievements Summary:

As illustrated in Figure *11*, after the well was connected to the production facility, Post fracturing flow back results for STG-5 at 46/64″ choke production performance shows almost 2 folds of increase in gas and 2.5 times the increase in gas rates. The wellhead pressure has also been increasing significantly from 200 psi to 600 psi as the well performance improved over time. The flow back for the post frac performed with energized CO2 was naturally without the need for coil tubing activation.

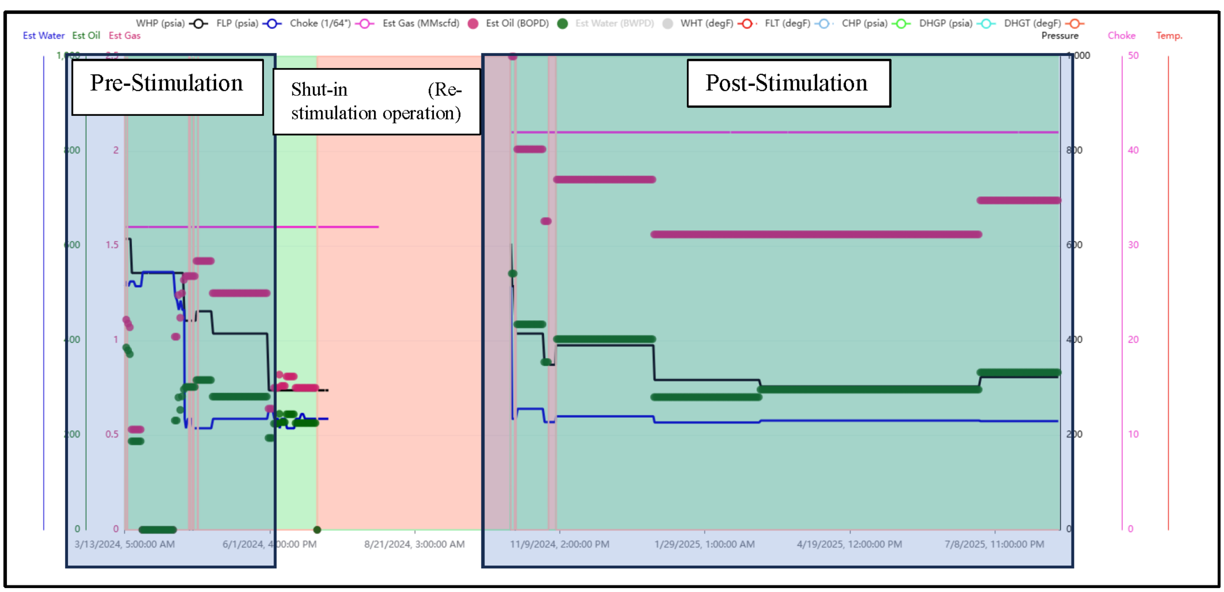


**Figure 11—Pre and Post Re-stimulation production results**

## Operational Lessons Learned – MSC Frac-ports Shifting Open/Close Operations

- Multiple function tests of shifting tool at various rates and pressures were essential to confirm operational readiness before entering the well.
- Shifting tool keys can suffer damage from high $H_2S/CO_2$ exposure and excessive mechanical loads. MPI (Magnetic Particle Inspection) is mandatory upon tool arrival from location and before re-dressing to detect cracking or deformation. Record emergency shear release values on the shifting tool before operations and never exceed 80% of that limit during shifting.
- A clean-out run to 60 ft below Stage 1 at the toe is critical to ensure sand/debris is removed and reduce risk of key damage or incomplete shifting. Avoid cleaning or milling beyond 60 ft below Stage 1 to prevent unnecessary tool wear.
- Increased drag and difficulty locating profiles in high-deviation completions require careful RIH management.
- Control RIH speed:100–150 ft/min generally.50–60 ft/min at least 500 ft before liner top, top packer, or liner hanger.
- Record pick-up/set-down weights every 3,000 ft for drag reference.
- Slow down when approaching target depth to capture accurate drag weights before shifting. Depth discrepancies of 20–40 ft between correlation runs can lead to missed latch attempts; repeated correlation tagging improved success rates.
- Soft tagging before applying full slack-off helped prevent accidental port activation or damage.
- Use smooth, slow movement (1 ft/15 sec if possible) to locate the profile—reducing shock load on keys.
- Maximum shifting load is 4,600 lbf straight push/pull (excluding drag). Without disciplined load control, there's a high chance of key bending, cracking, or premature shear.
- Avoid impact loads unless controlled through impact hammers or jars when required.
- Never stop circulation while keys of the shifting tool are engaged into the shifting profile of the sleeve to avoid sticking or partial engagement.
- Well exhibited continuous losses before and during the operation complicates confirmation of port status via returns.
- In high-loss scenarios, pressure monitoring + CT weight changes became more reliable indicators than flow returns.
- Recommend intermediate pressure testing after closing each set of sleeves to detect possible leaks early and re-verify closures.
- Ensure CT injector head maintenance and proper gripping before long runs; injector slippage caused repeated POOH/RIH cycles and added NPT.

## Conclusion

The successful deployment of advanced multistage completion systems has enabled rigless restimulation in the North Kuwait Jurassic Gas fields with minimum required intervention and demonstrated the value of integrating reservoir data, completion design, real-time diagnostics, and tailored stimulation techniques to optimize reservoir performance in mature, pressure-depleted carbonate reservoirs.

The introduction of energized $CO_2$ acid fracturing proved instrumental in enhancing stimulation effectiveness, particularly in selectively targeting low-productivity zones without compromising the integrity of previously stimulated intervals. The ability to mobilize treatment fluids with reduced water volumes and improved cleanup has significantly improved production rates while mitigating formation damage and flowback challenges. In addition, the lessons learned in mechanical sleeve performance under reversed differential pressure conditions prompted the redesign of sleeve systems, ensuring they meet the growing requirements for re-stimulation while preserving Monobore ID for future zonal isolation or plug-based interventions.

The case study's well demonstrated measurable improvements in post-stimulation oil and gas production, validating the integrated strategy adopted for selective re-stimulation. This approach provides a scalable and cost-effective solution to extend the productive life of Jurassic wells, supporting the importance of tailored completion technologies and cross-disciplinary collaboration in unlocking the remaining potential of tight carbonate plays.

## Acknowledgements

The authors of this paper would like to thank the management of Kuwait Oil Company for permission to publish this paper. Similarly, the authors would like to recognize all the personnel in the multi-disciplinary teams in KOC NKJG assets for their invaluable contribution to the success of the restimulation campaigns in openhole Multistage completions in North Kuwait.

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